Mattia Piccinini

TUM Global Post-doctoral Fellow in Autonomous Vehicle Systems

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LinkedIn | Research Gate | Google Scholar

Research Interests

My research lies at the intersection of **robotics**, **machine learning**, and **control** engineering. I develop algorithms for real-time trajectory planning, control, and estimation of autonomous robots, with a focus on **ground vehicles** in uncertain, dynamic environments. I like to integrate prior knowledge of the robot dynamics into learning-based methods, such as neural networks, to enhance their **generalization to unseen scenarios** with minimal training data.

Education

11/2019 – 04/2024
Ph.D. in Autonomous Systems, University of Trento, Italy
Doctorate awarded with highest distinction (*summa cum laude*)
Ph.D. thesis: *Artificial Drivers for Online Time-Optimal Vehicle Trajectory Planning and Control*
Supervisor: Prof. Francesco Biral

09/2017 – 10/2019
M.Sc. in Mechatronics Engineering, University of Trento, Italy
Final grade: 110/110 *summa cum laude*
GPA: 30/30, 80% exams with maximum score, 60% exams passed with honors
Thesis: *Path planning and control of self-driving vehicles at the limits of handling* [pdf].

09/2014 – 09/2017
B.Sc. in Industrial Engineering, University of Trento, Italy
Final grade: 110/110 *summa cum laude*
GPA: 30/30, 90% exams with maximum score, 60% exams passed with honors
Thesis: *Degradation of glycol-based cooling fluids causing corrosion in machine tools refrigeration circuits*. The thesis was carried out in cooperation with the companies Pama S.p.A. and Rittal GmbH.

09/2009 – 06/2014
Scientific High School Diploma
Liceo Scientifico A. Rosmini, Rovereto, Italy
Final grade: 100/100

Research Experience

12/2024 – present
TUM Global Post-doctoral Fellow
Technical University of Munich, Munich, Germany
Supervisor: Prof. Johannes Betz

04/2024 – 11/2024
Post-doctoral Researcher
University of Trento, Trento, Italy
Supervisor: Prof. Francesco Biral

03/2022 – 06/2022
Visiting Ph.D. Student
Universität der Bundeswehr, München, Germany
Supervisor: Prof. Matthias Gerdts

Teaching Experience

09/2021 – 02/2022
09/2020 – 02/2021
11/2019 – 02/2020

Teaching Assistant, EIT Digital Master School, University of Trento, Italy
Course of *Vehicle Dynamics, Planning and Control of Robotic Cars*.

Work Experience

09/2018 – 06/2019

Control and Simulation Engineer
Formula SAE Project, Dynamics and Modeling Team, University of Trento, Italy
- Development of vehicle multibody models to validate vehicle controllers.
- Parameter optimization for the mechatronic powertrain.

09/2017 – 07/2018

Mechanical Design Engineer
Formula SAE Project, Powertrain Design Team, University of Trento, Italy
- Responsible for the mechanical transmission unit.

06/2016 – 08/2017

Cooling System Engineer
Formula SAE Project, Cooling System Design Team, University of Trento, Italy
- Design and manufacturing of the cooling system for an electric FSAE vehicle.

08/2018 – 09/2018

Engineering Internship
Pama S.p.A., Rovereto (Italy)
- Design of innovative solutions for the kinematic chain in a large-scale, high-precision milling machine divider.

09/2016

Engineering Internship
Pama S.p.A., Rovereto (Italy)
- Experimental setup to investigate corrosion phenomena arising from the degradation of glycol-based cooling fluids within machine tool circuits.

02/2016
09/2015
02/2015

Engineering Internship
Pama S.p.A., Rovereto (Italy)
- Design and analysis of mechanical and hydraulic assemblies for large-scale high-precision milling and boring machine tools.

Publications and Patents

Peer-Reviewed Journal Papers

1. M. Piccinini, S. Taddei, M. Larcher, M. Piazza, and F. Biral, "A physics-driven artificial agent for online time-optimal vehicle motion planning and control," *IEEE Access*, vol. 11, pp. 46 344–46 372, 2023. DOI: 10.1109/ACCESS.2023.3274836.

2. M. D. Lio, M. Piccinini, and F. Biral, "Robust and sample-efficient estimation of vehicle lateral velocity using neural networks with explainable structure informed by kinematic principles," *IEEE Transactions on Intelligent Transportation Systems*, pp. 1–15, 2023. DOI: 10.1109/TITS.2023.3303776.

3. M. Piccinini, M. Larcher, E. Pagot, D. Piscini, L. Pasquato, and F. Biral, "A predictive neural hierarchical framework for on-line time-optimal motion planning and control of black-box vehicle models," *Vehicle System Dynamics*, vol. 61, no. 1, pp. 83–110, 2023. DOI: 10.1080/00423114.2022.2035776. eprint: https://doi.org/10.1080/00423114.2022.2035776.

4. E. Pagot, M. Piccinini, E. Bertolazzi, and F. Biral, "Fast planning and tracking of complex autonomous parking maneuvers with optimal control and pseudo-neural networks," *IEEE Access*, vol. 11, pp. 124 163–124 180, 2023. DOI: 10.1109/ACCESS.2023.3330431.

5. M. Piccinini, S. Taddei, E. Pagot, E. Bertolazzi, and F. Biral, "How optimal is the minimum-time manoeuvre of an artificial race driver?" *Vehicle System Dynamics*, vol. 0, no. 0, pp. 1–28, 2024. DOI: 10.1080/00423114.2024.2407176.

6. M. Piccinini, M. Zumerle, J. Betz, and G. P. R. Papini, "A road friction-aware anti-lock braking system based on model-structured neural networks," *IEEE Open Journal of Intelligent Transportation Systems*, pp. 1–1, 2025. DOI: 10.1109/OJITS.2025.3563347.

7. A. Langmann, S. Kohl, L. Ögretmen, M. Piccinini, and J. Betz, "A multi-stage time-variant motion planner for agile autonomous driving maneuvers," *IEEE Transactions on Intelligent Vehicles*, 2025, under review.

8. D. Wang, R. Brusnicki, Z. Lai, Y. Wu, M. Piccinini, R. Yang, W. Li, and J. Betz, "Extrospective prediction for autonomous driving in emergency cut-in scenarios," *IEEE Transactions on Intelligent Transportation Systems*, 2025, under review.

9. M. Piccinini, S. Gottschalk, M. Gerdts, and F. Biral, "Computationally efficient minimum-time motion primitives for vehicle trajectory planning," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 5, pp. 642–655, 2024. DOI: 10.1109/OJITS.2024.3476540.

Peer-Reviewed Conference Papers

1. E. Pagot, M. Piccinini, and F. Biral, "Real-time optimal control of an autonomous rc car with minimum-time maneuvers and a novel kineto-dynamical model," in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 2390–2396. DOI: 10.1109/IROS45743.2020.9340640.

2. M. Piccinini, S. Taddei, J. Betz, and F. Biral, "Kineto-dynamical planning and accurate execution of minimum-time maneuvers on three-dimensional circuits," in *IEEE International Conference on Robotics and Automation (ICRA)*, accepted, 2025. URL: https://arxiv.org/abs/2502.03454.

3. M. Kaufeld, M. Piccinini, and J. Betz, "Mp-rbfn: Learning-based vehicle motion primitives using radial basis function networks," in *IEEE Intelligent Transportation Systems Conference (ITSC)*, under review, 2025.

4. Y. Gao, M. Piccinini, K. Moller, A. Alanwar and J. Betz, "From words to collisions: Llm-guided evaluation and adversarial generation of safety-critical driving scenarios," in *IEEE Intelligent Transportation Systems Conference (ITSC)*, under review, 2025.

5. M. Piccinini, A. Mungiello, G. Jank, G. P. R. Papini, and J. Betz, "Model-structured neural networks to control the steering dynamics of autonomous race cars," in *IEEE Intelligent Transportation Systems Conference (ITSC)*, under review, 2025.

6. F. Werner, S. Sagmeister, M. Piccinini, and J. Betz, "A quasi-steady-state black box simulation approach for the generation of g-g-g-v diagrams," in *IEEE Intelligent Vehicles Symposium (IV)*, accepted, 2025.

7. B. Zhou, B. Zarrouki, M. Piccinini, C. Hu, L. Xie, and J. Betz, "Safe reinforcement learning with a predictive safety filter for motion planning and control: A drifting vehicle example," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, under review, 2025.

8. S. Taddei, M. Piccinini, and F. Biral, "Biasing the driving style of an artificial race driver for online time-optimal maneuver planning," in *IEEE Intelligent Vehicles Symposium (IV)*, accepted, 2025.

9. S. Gottschalk, M. Gerdts, and M. Piccinini, "Reinforcement learning and optimal control: A hybrid collision avoidance approach," in *Proceedings of the 10th International Conference on Vehicle Technology and Intelligent Transport Systems - VEHITS*, INSTICC, SciTePress, 2024, pp. 76–87, ISBN: 978-989-758-703-0. DOI: 10.5220/0012569800003702.

10. M. Piazza, M. Piccinini, S. Taddei, and F. Biral, "Mptree: A sampling-based vehicle motion planner for real-time obstacle avoidance," 10, 17th IFAC Symposium on Control of Transportation Systems CTS 2024, vol. 58, 2024, pp. 146–153. DOI: https://doi.org/10.1016/j.ifacol.2024.07.332.

11. M. Piccinini, S. Taddei, M. Piazza, and F. Biral, "Impacts of g-g-v constraints formulations on online minimum-time vehicle trajectory planning," 10, 17th IFAC Symposium on Control of Transportation Systems CTS 2024, vol. 58, 2024, pp. 87–93. DOI: https://doi.org/10.1016/j.ifacol.2024.07.323.

Peer-Reviewed Workshop Papers

1. E. Pagot, M. Piccinini, A. Plebe, E. Bertolazzi, and F. Biral, "Real-time autonomous parking in unstructured scenarios with an indirect optimal control approach," in *Workshop Behavior-Driven Autonomous Driving in Unstructured Environments (BADUE), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022. URL: https://obj.umiacs.umd.edu/badue-accepted/8.pdf.

2. M. Piccinini, S. Gottschalk, M. Gerdts, and F. Biral, "Neural motion primitives for online time-optimal vehicle trajectory planning," in *Conference AUTOMATICA.IT*, 2024.

3. S. Taddei, M. Piccinini, E. Pagot, and F. Biral, "Artificial racing coach: Teaching humans how to maximize a racing vehicle's performance and drive at its limits," in *Conference AUTOMATICA.IT*, 2024.

4. M. Piccinini, S. Taddei, M. Larcher, M. Piazza, and F. Biral, "A physics-driven framework for online minimum-time vehicle motion planning and control," in *Conference AUTOMATICA.IT*, 2023.

Other Publications

1. M. Piccinini and S. Mariano, "Localization and control of autonomous vehicles in an urban intersection scenario," 2019. URL: https://www.researchgate.net/publication/334945954_Localization_and_Control_of_Autonomous_Vehicles_in_an_Urban_Intersection_Scenario.

Invited Talks, Presentations and Poster Sessions

1. M. Piccinini, E. Pagot, and S. Taddei, "Virtual Driver to Control Race Cars near the Limits", *PhDII Poster Session*, University of Trento, 2022.

2. M. Piazza and M. Piccinini, "MPTree: Motion Primitive Tree Exploration for Trajectory Planning with Dynamic Obstacle Avoidance", *Poster Session at the Industrial Engineering Day*, University of Trento, 2022.

3. M. Piccinini, S. Taddei, and F. Biral, "Artificial Race Driver", *Poster Session at the Industrial Engineering Day*, University of Trento, 2023.

4. M. Piccinini, "Virtual race drivers for planning and control with robotic cars", *Seminar at the Engineering Mathematics Days*, Universität der Bundeswehr, Munich, 2022.

5. M. Piccinini, A. Mungiello, J. Betz, and G. P. R. Papini, "Model-Structured Neural Networks to Model & Control Robots", *Late-Breaking Results Poster Session*, International Conference on Robotics and Automation (ICRA), Atlanta, USA, 2025.

6. M. Piccinini, "Artificial Drivers to Learn the Vehicle Dynamics, Plan and Execute Online Time-Optimal Maneuvers", *Seminar*, Technical University of Munich, 2024.

7. M. Piccinini and G. P. R. Papini, "Model-Structured Neural Networks to Model and Control Physical Systems", *Seminar for PhD students*, University of Trento, 2024.

8. M. Piccinini, "Model-Structured Neural Networks to Model and Control Vehicles and Robots", *Doktorandenkolloquium Seminar*, Technical University of Munich, Berchtesgaden, 2025.

Patents

1. M. D. Lio, F. Biral, and M. Piccinini, "Device and method, based on neural networks, for estimating the lateral speed of vehicles", WO2023166536A1, 2023. URL: https://patents.google.com/patent/WO2023166536A1/en?oq=PCT%2fIT2023%2f050058.

Awards

03/2025 **EuroTech Visiting Researcher Award**
15-day research travel grant to collaborate with the Eindhoven University of Technology (Prof. Mauro Salazar), following a competitive selection of a research proposal.

12/2024 **TUM Global Post-doctoral Fellowship**
2-year funding for early-career researchers, following a positive evaluation of a research proposal (**10% success rate**).

07/2024 **IFAC Young Author Award**
Awarded at the 17th IFAC Symposium on Control in Transportation Systems, Cyprus.

05/2024 **Nomination for best paper award**
10th International Conference on Vehicle Technology and Intelligent Transport Systems (VEHITS).

04/2024 **Ph.D. graduation with highest distinction** (*summa cum laude*)

11/2023 **Best Poster Award**
Industrial Engineering Day, University of Trento (Italy), 2023

10/2019 **M.Sc. graduation with highest distinction** (*summa cum laude*)

09/2017 **B.Sc. graduation with highest distinction** (*summa cum laude*)

Academic Service

Peer-Review Activities

Journals

- IEEE Transactions on Intelligent Transportation Systems
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Mechatronics
- IEEE Transactions on Industrial Electronics
- IEEE Transactions on Control Systems Technology
- IEEE Transactions on Robotics
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Access
- Vehicle System Dynamics
- Springer Nature - International Journal of Machine Learning and Cybernetics
- Springer Optimization and Engineering
- Mechanical Systems and Signal Processing
- MDPI Vehicles, MDPI Machines, MDPI Electronics
- MDPI Drones, MDPI Symmetry
- MDPI World Electric Vehicle Journal

Conferences

- IEEE International Conference on Robotics and Automation (ICRA)

- IEEE Intelligent Vehicles Symposium (IV)
- IEEE Conference on Decision and Control (CDC)
- IEEE Conference on Control Technology and Applications
- IEEE International Conference on Real-time Computing and Robotics
- IEEE International Conference on Advanced Motion Control
- International Conference on System Theory, Control and Computing (ICSTCC)
- IFAC Symposium on Control in Transportation Systems

Conference Committees

- **Chair** of the session "Path Planning", at the Automatica.it conference, Bolzano (Italy), 2024
- **Co-chair** of the session "Modeling, Control, and Optimization of Transportation Systems", at the 17th IFAC Symposium on Control in Transportation Systems (CTS), Ayia Napa (Cyprus), 2024

Supervised Theses and Projects

- 2025-present, Technical University of Munich (Germany): 2 Master Students, 2 Visiting PhD Student, 2 PhD Students
- 2019-present, University of Trento (Italy): 8 Master Students, 1 PhD Student
- 2021-2022, EIT Digital Master School (EU): 1 Master Student, 1 Term Project
- 2020-2021, HAN University of Applied Sciences (Netherlands): 1 Master Student

Skills

Programming	Matlab, Simulink, Python, C/C++, Maple, Mathematica, Ruby
Machine Learning Frameworks	Tensorflow, Keras, PyTorch
Development Tools	GitHub, GitLab, Bitbucket
CAD Software	SolidWorks, Inventor, AutoCAD
Operating Systems	MacOS, Windows, Linux
Other Computer Skills	LaTeX, Microsoft Word, Microsoft Excel, Microsoft Power-Point, Apple Keynote, Apple Pages
Others	Driving licence B
Languages	English: full professional proficiency (Cambridge Certificate of Advanced English, level C1) German: upper-intermediate knowledge Italian: native

Personal Details and Interests

- Date of birth: 16/07/1995
- Nationality: Italian
- Personal interests: traveling, swimming, hiking

References

- Prof. Johannes Betz, Technische Universität München, München (Germany),
  phone: +49 89 289 10407, email: johannes.betz@tum.de

- Prof. Francesco Biral, University of Trento, Trento (Italy),
  phone: +39 (0)461 282513, email: francesco.biral@unitn.it
- Prof. Matthias Gerdts, Universität der Bundeswehr, München (Germany),
  phone: +49 (0)89 6004 2672, email: matthias.gerdts@unibw.de
- Prof. Mauro Da Lio, University of Trento, Trento (Italy),
  phone: +39 (0)461 282501, email: mauro.dalio@unitn.it
- Prof. Gastone Pietro Rosati Papini, University of Trento, Trento (Italy),
  phone: +39 (0)461 285352, email: gastone.rosatipapini@unitn.it

Munich, 04/05/2025

Mattia Piccinini